

Asha Jyothi Boddu

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Education

BVRIT Hyderabad College of Engineering for Women

B.Tech – Computer Science in Artificial Intelligence and Machine Learning

2023 – 2027

Hyderabad, India · **CGPA: 8.80/10**

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Vite, Tailwind CSS, TanStack Query, FastAPI, Flask, Pydantic, Pandera, NumPy, Pandas, scikit-learn, XGBoost, TensorFlow, OpenCV, MediaPipe, SQLAlchemy

AI & LLM: LLM tool-calling agents, RAG, Model Context Protocol (MCP), Groq, NVIDIA NIM, OpenRouter, conformal prediction, Isolation Forest, Prophet, Chronos, Jev, Laya, evaluation harnesses

DevOps, Testing & Data: Git, GitHub Actions (CI/CD), CodeQL, Dependabot, OpenSSF Scorecard, Docker, Render, pytest, Hypothesis, mutmut (mutation testing), ruff, PostgreSQL, MySQL, SQLite

Concepts: DSA, OOP, Distributed Systems, LLM Evaluation & Guardrails, Provider Failover, Hybrid Retrieval, UPI Payments

Internship & Mentorship

AptPath – AI/ML Development Intern

July 2025 – August 2025 · Hyderabad, India

- Built **YogaAlign** – a yoga pose classification system (Python, Flask, OpenCV, MediaPipe) achieving **92% accuracy** across 15 pose categories, and cut real-time frame latency **30%** [repo]

Ananya's Atlassian Mentorship Program – Mentee, DSA and Backend Track

November 2025 – May 2026

- Selected for a **6-month structured mentorship** in scalable distributed system design; applied OOP principles and collaborative Git workflows across guided multi-service projects

Projects

AT&C Loss Detection and Inspection Intelligence System [live] [repo]

May 2026

Python, FastAPI, scikit-learn, Pandas, React, TypeScript, LLM agents (Groq, NVIDIA NIM), GitHub Actions

- Built a 4-layer theft-detection pipeline (transformer energy balance, own-history baseline, peer comparison, Isolation Forest) ranking inspections by rupee recovery; on 20 unseen simulated networks: **0.82 precision, 0.93 recall, F1 0.87**, recall gated in CI
- Added a tool-calling **copilot**, **RCA inspection-brief agent** and **rule-driven smart alerts** (Groq with NIM failover and rule-based fallback; every number from read-only tools) with an **evals harness** (0 invented meter ids), an LLM **observability** page, and an **MCP server** exposing the 8 read-only tools to any MCP client; deployed live on Render
- Checked the simulator against real London smart-meter data (realism score 61%→84%) and benchmarked forecasters on real feeders, replacing the old model (MASE 1.52→0.83)

Retry Budget Allocator [live] [repo]

September 2026

Python, FastAPI, Pydantic, React, Pandera, mutmut, GitHub Actions, LLM-based reasoning generation

- Designed a cause-aware engine allocating a regulated 3-attempt UPI AutoPay retry budget: **50% fewer attempts** than a fixed schedule (70 vs. 141) with **0 wasted attempts** and **zero compliance violations**; reported where it loses (30 vs. 35 recoveries; wins on money only above Rs 147.82/attempt) across a 27-setting sweep and 10 seeds
- Kept the LLM off the decision path (deterministic classify → prioritize → decide, enforced by an AST test); an outage degrades wording, never decisions
- Built the eval and CI/CD pipeline: Pandera data contract, **mutation testing** (31/32 mutants killed), property-based fuzzing (1200+ cases), CodeQL, and eval gates that fail the build on any violation
- Shipped a live Render dashboard (landing, guided tour, evals view); the model writes only customer text, with a template fallback

HealthAI – Triage Agent with Safety Evals [live] [repo]

August 2026

Python, FastAPI, React, RAG (model2vec), Groq/NVIDIA NIM tool-calling, SQLite, GitHub Actions

- Built a tool-calling triage agent over a RAG knowledge base (hybrid dense + keyword retrieval) with provider failover; deterministic red-flag rules **can only raise a level**, so a model error cannot downgrade an emergency
- Wrote a 60-case **eval harness** gated in CI on missed emergencies (rules-only baseline: 100% of red-flag cases caught, 85% of all emergencies; the 2 misses are reported) plus an Ops page logging every model call; deployed live with an in-app tour of every page
- Added an optional conformal-prediction second opinion (escalate-only) behind a provider-agnostic adapter for external decision-model endpoints (e.g., Jev, Laya), and 135 tests with a fake LLM transport, so CI never needs a model key

Research & Achievements

- Finalist – Code by Groww 2026:** one of 9 finalists out of 2,900+ registrations (top ~0.3%) [repo]
- Flipkart GRiD 8.0 Round 3 Qualifier (2026):** top 0.5% nationwide of 400,000+ applicants
- CodeChef 3-Star Rated** – solved **850+ problems** across LeetCode, CodeChef, and Codeforces
- Presented paper **Significance of Emergent Technologies in Teaching Learning Processes** at ICIARD 2024, Metarath University, Thailand (February 26–28, 2024)

Leadership & Activities

Founding Member & Joint Secretary – A.S.P.I.R.E

2024 – Present

First Student-led AI Club, BVRIT Hyderabad

- Co-founded BVRIT Hyderabad's first AI/ML student club; led workshops reaching **100+ students**; organized **9 events** including an inter-college competition with **200+ participants** (2025, 2026)

Aspire Leaders Program – Aspire Institute (Harvard-affiliated), Global Leadership Development

Completed March 2025